

Integrating RAG-Enhanced AI to Support Cognitive, Affective, and Psychomotor Balance in K–12 Lesson Plans

Abstract

Although Social-Emotional Learning (SEL) is widely recognized as essential to whole-child development, affective objectives remain underrepresented in K–12 lesson design. This study investigates the use of a Retrieval-Augmented Generation (RAG)–enhanced AI system to strengthen SEL integration and rebalance instructional goals, aligned with Taiwan’s 108 Curriculum Guidelines. Using Bloom’s Taxonomy, the CASEL framework, and psychomotor learning theory, we analyzed 58 lesson plans by pre-service teachers. A transformer-based classifier revealed only 17% of objectives targeted the affective domain. After AI-driven augmentation, SEL coverage rose to 32%, with significant gains in self-management and decision-making. This research presents a scalable, explainable AI framework for embedding SEL into curriculum planning—advancing emotional depth, equity, and teacher agency in pre-service teacher education.

Keyword: Social-Emotional Learning (SEL), Retrieval-Augmented Generation (RAG), Taiwan 108 Curriculum Guidelines, Bloom’s Taxonomy

1. Objectives or purposes

As 21st-century education continues to evolve in response to rapid technological change and increasingly complex social demands, the call for holistic and domain-balanced instructional design has grown more urgent. While frameworks such as Bloom’s Taxonomy (Bloom & Krathwohl, 1956) and the CASEL competencies (Durlak et al., 2011; Lawson et al., 2019) have long emphasized the need to integrate cognitive, affective, and psychomotor learning, empirical research reveals that existing K–12 lesson plans remain disproportionately focused on lower-order cognitive objectives (Muhayimana et al., 2022). Social-Emotional Learning (SEL), though widely recognized as critical to student development, often remains under-implemented in daily classroom practice due to limited teacher support and lack of systematized integration strategies (Wu et al., 2019).

At the same time, recent advances in artificial intelligence—particularly the use of Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) frameworks—offer new possibilities for scaffolding instructional design and rebalancing curriculum emphasis through automated yet explainable pedagogical augmentation (Rutherford et al., 2025; Swacha & Gracel, 2025). However, their application in the context of domain-level curriculum balance remains underexplored and under-evaluated, especially with regard to teacher autonomy and pedagogical transparency.

To address these gaps, this study develops and evaluates an explainable, RAG-enhanced AI system that analyzes and revises K–12 lesson plans through the

lens of Bloom's Taxonomy, CASEL's SEL framework, and psychomotor learning domains. The goal is to create an automated yet teacher-controllable framework that diagnoses domain imbalances and proposes evidence-aligned, cognitively and emotionally enriching enhancements.

- **RQ1:** Based on classifications from Bloom's Taxonomy, CASEL, and psychomotor learning frameworks, what is the distribution of cognitive, affective, and psychomotor domain representation in current K–12 lesson plans??
- **RQ2:** Can a RAG-enhanced AI system significantly increase the proportion of affective (SEL) objectives and higher-order cognitive objectives in existing lesson plans?
- **RQ3:** To what extent do RAG-enhanced lesson plans align more closely with 21st-century competency frameworks than the original versions?

2. Theoretical framework

This study integrates multiple educational theories to support the design and evaluation of a RAG-enhanced AI system for rebalancing K–12 lesson plans across cognitive, affective, and psychomotor domains. The framework is grounded in Bloom's Taxonomy for cognitive development, the CASEL social-emotional learning competencies for affective growth, Simpson's psychomotor taxonomy for embodied learning, 21st-century competency frameworks (including Taiwan's 108 Curriculum Guidelines), and knowledge-building pedagogy rooted in social constructivist theory. These theories collectively inform both the analytical criteria for diagnosing domain imbalance and the instructional logic underlying AI-generated enhancements.

2-1. Cognitive Domain: Bloom's Taxonomy

Bloom's Taxonomy provides a foundational structure for classifying cognitive learning objectives along a hierarchical continuum—from lower-order skills such as remembering and understanding to higher-order skills including analyzing, evaluating, and creating (Arievitch, 2020). The Revised Taxonomy (Noor et al., 2020) emphasizes active engagement with content and has become a widespread tool for instructional design and assessment. Empirical analysis of Rwandan national exams revealed a strong overrepresentation of low-level cognitive tasks, with rote recall dominating over critical thinking across six years of high-stakes assessments (Muhayimana et al., 2022). Within this study, Bloom's framework serves as both an analytic rubric (for measuring imbalance in baseline lesson plans) and a target structure for AI-driven revisions, which are designed to increase higher-order thinking opportunities.

2-2. Affective Domain: CASEL Social-Emotional Learning Competencies

To capture the affective dimension of learning, this study adopts the CASEL framework, which defines five core social-emotional learning (SEL) competencies: self-awareness, self-management, social awareness, relationship skills, and responsible decision-making (Durlak et al., 2011; Taylor et al., 2017). SEL integration has been shown to yield improvements in student behavior, well-being, and academic

performance, especially when embedded across curricula (Mahoney et al., 2021). In the present study, lesson objectives are classified according to CASEL categories, and the AI system is tasked with augmenting lessons to better reflect SEL priorities—thereby rebalancing the domain representation of instructional content.

2-3. Psychomotor Domain: Simpson's Taxonomy

Although often underrepresented in general education curricula, the psychomotor domain remains essential for a holistic learning model. Simpson's taxonomy (1972) outlines seven levels of psychomotor development, including perception, set, guided response, mechanism, complex overt response, adaptation, and origination. A recent randomized-controlled trial in physical education demonstrated that integrating structured psychomotor activities significantly enhances students' procedural fluency and executive functioning, supporting the role of kinesthetic engagement in holistic learning (Invernizzi et al., 2022). This study incorporates the psychomotor domain by comparing linear and nonlinear pedagogies in physical education, demonstrating that movement-based instruction—emphasizing motor planning, creativity, and reflection—can significantly enhance executive function and procedural fluency, underscoring the importance of embodied learning in early education.

2-4. Curriculum Alignment: 21st-Century Competencies and Taiwan's 108 Curriculum Guidelines

Globally, educational policy has shifted toward cultivating 21st-century competencies such as creativity, critical thinking, communication, collaboration, and self-regulation (Voogt & Roblin, 2012). Taiwan's 108 Curriculum Guidelines reflect this shift through three overarching domains of core competencies: spontaneity (self-directed action), communication and interaction, and social participation (Lin & Chen, 2024). These curricular goals extend beyond academic knowledge to emphasize lifelong learning, civic responsibility, and personal well-being. In this study, these competency frameworks provide a criterion for evaluating RAG-enhanced lesson plans (RQ3), allowing us to assess whether the AI system helps align instructional content with future-oriented, nationally endorsed learning objectives.

3. Methods/ Data sources

3-1 background

We conducted this study within a university-level teacher training course in Taiwan, designed to prepare pre-service teachers for K–12 instructional practice. The course focused on curriculum design and pedagogical innovation, requiring each participant to develop original lesson plans aligned with Taiwan's 108 Curriculum Guidelines. As part of the course requirement, students created lesson plans spanning a range of subjects, including English, Chinese, mathematics, science, and integrated thematic units. The course served as a structured environment for participants to experiment with instructional design principles, supported by ongoing faculty guidance and peer collaboration.

A total of 58 lesson plans were collected as part of this research. These plans were used to diagnose domain imbalance across cognitive, affective, and psychomotor learning objectives, and to test the effectiveness of an AI-assisted system for enhancing domain coverage and curriculum alignment. with continuously designing and K–12 lesson plans. The submitted lesson plans spanned various subject areas including English, Chinese, mathematics, science, and integrated activities, offering a representative sample of Taiwan's curriculum design. Each plan included instructional goals, learning activities, and assessment strategies.

3-2 Data Collection and Analysis

The following data collection and analysis procedures were designed to address the three research questions concerning domain imbalance, AI-enhanced instructional design, and curriculum alignment within K–12 lesson plans developed by pre-service teachers.

RQ1: Domain Imbalance Diagnosis

Fifty-eight lesson plans were selected from participants' final submissions. Each learning objective within these plans was classified using a hybrid three-domain schema aligned with Taiwan's 108 Curriculum Guidelines:

- **Cognitive domain:** Mapped to Bloom's Taxonomy.
- **Affective domain:** Aligned with CASEL's five SEL competencies.
- **Psychomotor domain:** Based on Atkinson and Murrell (2019), representing hands-on and embodied learning activities.

Classification was performed by a transformer-based classifier (fine-tuned RoBERTa), trained on 800 manually coded teaching objectives. Each objective received:

- A primary domain label: Cognitive (C), Affective (A), or Psychomotor (P)
- A subdomain tag: Bloom level, CASEL competency, or psychomotor skill category

Human coders double-annotated 30% of the data (inter-rater reliability: $\kappa = 0.83$). Descriptive statistics were used to identify domain imbalance.

RQ2: RAG-Enhanced Plan Generation

Lesson plans with underrepresented domains were passed through a custom Retrieval-Augmented Generation (RAG) pipeline. This system:

- Diagnosed domain gaps via classifier output
- Retrieved relevant content fragments from a curated database of teacher-verified instructional activities
- Prompted a fine-tuned GPT-based LLM to generate new or revised objectives targeting missing domains

Example RAG prompt:

"Generate an activity aligned with the affective domain (empathy and self-management) for a Grade 8 science lesson about environmental responsibility."

Revised lesson plans were re-classified, and pre/post comparisons were analyzed via paired-sample t-tests for domain proportions. Effect sizes (Cohen's d) were reported.

RQ3: Curriculum Alignment Evaluation

Both original and RAG-enhanced lesson plans were scored using a rubric based on:

- Taiwan's 108 Curriculum Guidelines
- Voogt & Roblin's (2012) international 21st-century competencies

Nine core competencies were evaluated on a 3-point scale (0 = not present, 1 = implicit, 2 = explicit). Two curriculum experts rated all plans (inter-rater reliability $\kappa = 0.87$). Paired t-tests compared mean alignment scores.

3-2.1 Explainability and Teacher Control

To ensure transparency and preserve teacher agency, the AI system was designed with explainable features at multiple stages. Explainability is achieved through traceable retrieval paths, interpretable domain tagging, and editable outputs. For each enhancement, the system identifies instructional gaps using a domain classifier and retrieves relevant activities from a curated content base. These retrievals, along with structured prompts, are presented to the teacher for review. Suggested revisions are annotated with domain labels (e.g., CASEL competencies or Bloom's levels), and teachers can accept, reject, or modify them based on professional judgment. This human-in-the-loop design supports pedagogical oversight while leveraging the generative capabilities of RAG-enhanced AI.

3-3 Ethical Considerations

All lesson plans were anonymized. Participant consent was obtained for survey and activity data analysis. AI-generated content was reviewed for instructional appropriateness and ethical compliance under the 108 Curriculum Guidelines.

4. Results and conclusions

Regarding RQ1, 50% of instructional objectives were classified as cognitive, 17% as affective, and 33% as psychomotor. This indicates a strong emphasis on knowledge-based objectives and a relative neglect of social-emotional or hands-on learning components. Figure 1 presents this distribution in a pie chart format, clearly highlighting the underrepresentation of the affective domain.

[figure1]

Regarding RQ2, There was a significant increase in the proportion of affective objectives (from 17% to 32%, $t = 7.82$, $p < .01$) and an upward shift in cognitive complexity, with more objectives classified under Bloom's "Analyze," "Evaluate," and "Create" levels. Figure 2 shows these shifts visually in a grouped bar chart comparing pre- and post-enhancement domain proportions.

[figure2]

Regarding RQ3, figure 3 shows the average alignment scores across nine core competencies derived from Taiwan's 108 Curriculum Guidelines and international 21st-century frameworks. After RAG enhancement, average competency alignment scores improved significantly across all categories (e.g., creativity, critical thinking, ethics). A paired-sample t-test confirmed the significance of this improvement ($p < .01$ across all nine domains).

[figure3]

5. Scientific significance of the study

This study contributes to the growing field of AI-enhanced educational design by demonstrating how RAG can be used to rebalance instructional objectives and scaffold teacher development in alignment with national curriculum policies. Unlike many prior studies that focus solely on cognitive learning gains or task automation, this research integrates domain-level diagnostics, curriculum policy (Taiwan's 108 Guidelines), and explainable AI to create a replicable instructional design framework. The use of transformer-based classification and localized RAG pipelines offers a scalable solution to long-standing imbalances in lesson planning—particularly the underrepresentation of affective and psychomotor goals. The methodology and findings serve as a prototype for integrating large language models in curriculum-supportive, teacher-centric, and culturally aligned ways, contributing both theoretically and practically to research in educational technology and teacher professional development.

6. Limitations and Future Directions

This study is limited by its sample size of 58 pre-service teacher lesson plans and its focus on Taiwan's 108 Curriculum Guidelines, which may affect generalizability to other contexts. Additionally, while the AI system emphasizes explainability and teacher control, user perceptions of these features were not evaluated. Future research should examine broader applications and explore how teachers interact with and interpret AI-generated enhancements in practice.

Reference

- Bloom, B. S., & Krathwohl, D. R. (1956). Taxonomy of educational objectives; the classification of educational goals by a committee of college and university examiners. Handbook I: Cognitive Domain. New York, NY; Longmans, Green.
- Durlak, J. A., Weissberg, R. P., Dymnicki, A. B., Taylor, R. D., & Schellinger, K. B. (2011). The Impact of Enhancing Students' social and Emotional Learning: A Meta-Analysis of School-Based Universal Interventions. *Child Development*,

- 82(1), 405–432. <https://doi.org/10.1111/j.1467-8624.2010.01564.x>
- Lawson, G. M., McKenzie, M. E., Becker, K. D., Selby, L., & Hoover, S. A. (2018). The core components of Evidence-Based Social Emotional Learning programs. *Prevention Science*, 20(4), 457–467. <https://doi.org/10.1007/s11121-018-0953-y>
- Muhayimana, T., Kwizera, L., & Nyirahabimana, M. R. (2022). Using Bloom's taxonomy to evaluate the cognitive levels of Primary Leaving English Exam questions in Rwandan schools. *Curriculum Perspectives*, 42(1), 51–63. <https://doi.org/10.1007/s41297-021-00156-2>
- Wu, W., Kao, H., Wu, S., & Wei, C. (2019). Development and evaluation of affective domain using student's feedback in entrepreneurial massive open online courses. *Frontiers in Psychology*, 10. <https://doi.org/10.3389/fpsyg.2019.01109>
- Rutherford, T., Rodrigues, A., Duque-Baird, S., Veng, S., Mykyta-Chomsky, R., Cao, Y., Chisholm, K., & Bergwall, E. (2025). "I just think it is the way of the future": Teachers use of ChatGPT to Develop Motivationally-Supportive Math Lessons. *Computers and Education Artificial Intelligence*, 100367. <https://doi.org/10.1016/j.caeai.2025.100367>
- Swacha, J., & Gracel, M. (2025). Retrieval-Augmented Generation (RAG) chatbots for Education: A survey of applications. *Applied Sciences*, 15(8), 4234. <https://doi.org/10.3390/app15084234>
- Arievitch, I. M. (2020). The vision of Developmental Teaching and Learning and Bloom's Taxonomy of educational objectives. *Learning Culture and Social Interaction*, 25, 100274. <https://doi.org/10.1016/j.lcsi.2019.01.007>
- Noor, N. a. M., Saim, N. M., Alias, R., & Rosli, S. H. (2020). Students' performance on cognitive, psychomotor and affective domain in the course outcome for embedded course. *Universal Journal of Educational Research*, 8(8), 3469–3474. <https://doi.org/10.13189/ujer.2020.080821>
- Muhayimana, T., Kwizera, L., & Nyirahabimana, M. R. (2022b). Using Bloom's taxonomy to evaluate the cognitive levels of Primary Leaving English Exam questions in Rwandan schools. *Curriculum Perspectives*, 42(1), 51–63. <https://doi.org/10.1007/s41297-021-00156-2>
- Taylor, R. D., Oberle, E., Durlak, J. A., & Weissberg, R. P. (2017). Promoting Positive Youth Development Through School-Based Social and Emotional Learning Interventions: A Meta-Analysis of Follow-Up Effects. *Child Development*, 88(4), 1156–1171. <https://doi.org/10.1111/cdev.12864>
- Mahoney, J. L., Weissberg, R. P., Greenberg, M. T., Dusenbury, L., Jagers, R. J., Niemi, K., Schlinger, M., Schlund, J., Shriver, T. P., VanAusdal, K., & Yoder, N. (2020). Systemic social and emotional learning: Promoting educational success for all preschool to high school students. *American Psychologist*, 76(7), 1128–1142. <https://doi.org/10.1037/amp0000701>

Simpson, E. (1972) *The Classification of Educational Objectives in the Psychomotor Domain*. Gryphon House, Washington DC.

Invernizzi, P. L., Signorini, G., Rigon, M., Larion, A., Raiola, G., D'Elia, F., Bosio, A., & Scurati, R. (2022). Promoting Children's Psychomotor Development with Multi-Teaching Didactics. *International Journal of Environmental Research and Public Health*, 19(17), 10939. <https://doi.org/10.3390/ijerph191710939>

Voogt, J., & Roblin, N. P. (2012). A comparative analysis of international frameworks for 21st century competences: Implications for national curriculum policies. *Journal of Curriculum Studies*, 44(3), 299–321. <https://doi.org/10.1080/00220272.2012.668938>

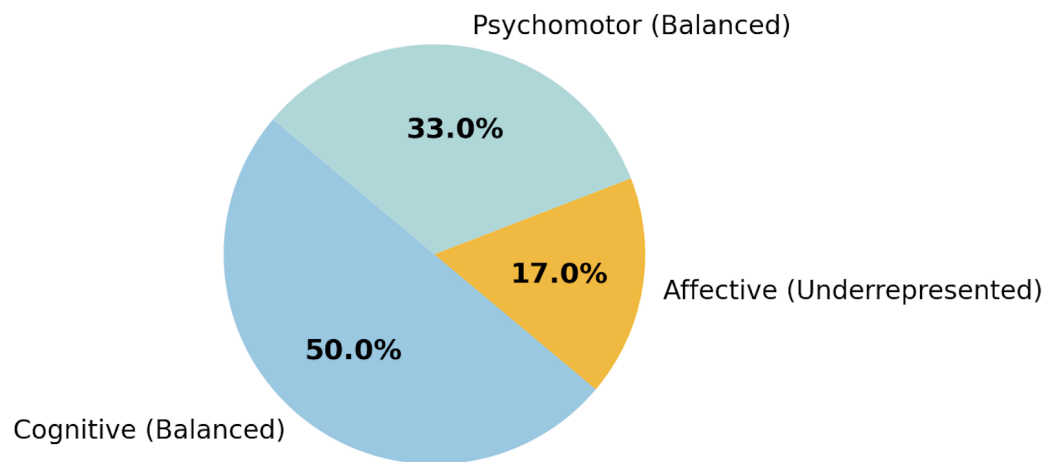


Figure1 Domain Distribution

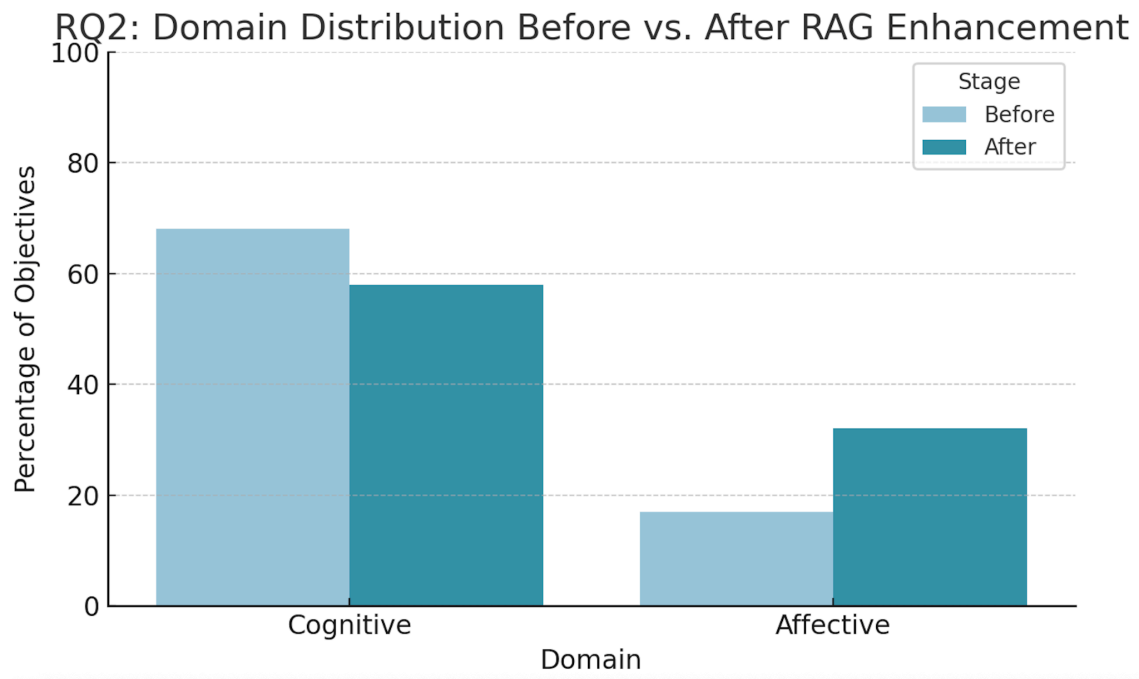


Figure2 pre- and post-enhancement domain proportions.

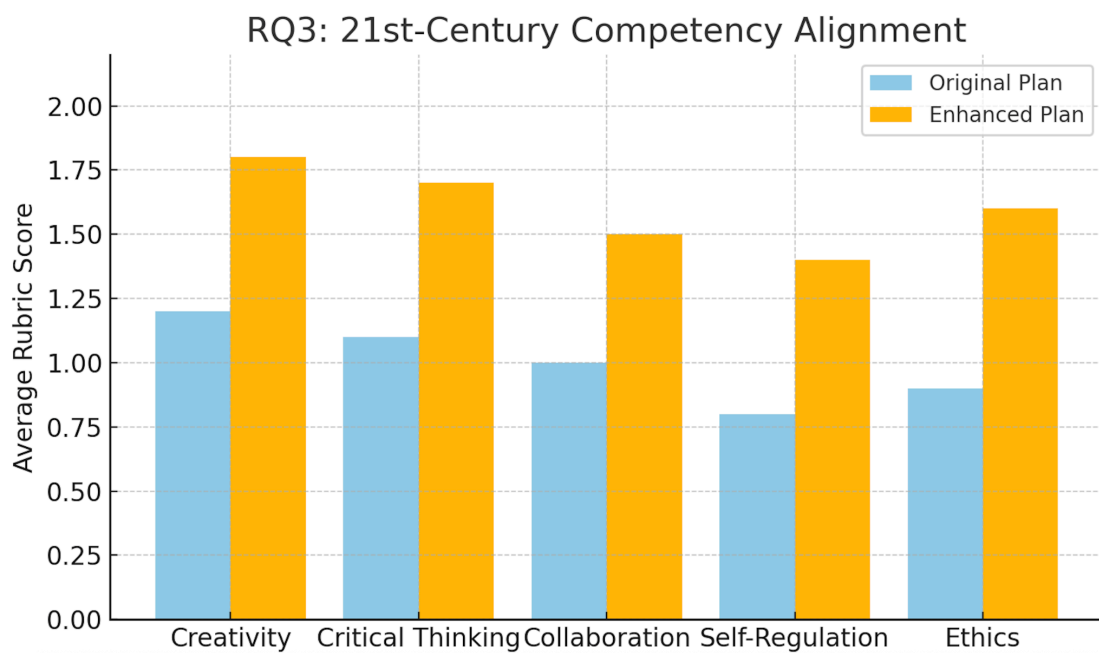


Figure3 Competency alignment bar chart